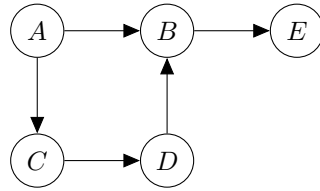


CS 791: Quiz 2

1. Given a BN:-



$$\Pr(A = 0) = 0.5$$

$$\Pr(B = 0 \mid A = 0, D = 0) = 0.1,$$

$$\Pr(B = 0 \mid A = 0, D = 1) = 0.2$$

$$\Pr(B = 0 \mid A = 1, D = 0) = 0.6,$$

$$\Pr(B = 0 \mid A = 1, D = 1) = 0.9$$

$$\Pr(C = 0 \mid A = 0) = 0.3,$$

$$\Pr(C = 0 \mid A = 1) = 0.4$$

$$\Pr(D = 0 \mid C = 0) = 0.2,$$

$$\Pr(D = 0 \mid C = 1) = 0.6$$

$$\Pr(E = 0 \mid B = 0) = 0.7,$$

$$\Pr(E = 0 \mid B = 1) = 0.8$$

(a) Which of the following CIs is/are correct?

- i. $A \perp\!\!\!\perp D \mid B$
- ii. $A \perp\!\!\!\perp D \mid C$
- iii. $A \perp\!\!\!\perp D \mid B, C$
- iv. $C \perp\!\!\!\perp E \mid B$

..2 (multicorrect, -0.5 for each wrong answer)

- i. The path $A-C-D$ is not blocked
- ii. The path $A-C-D$ is blocked by C and $A-B-D$ is an immorality
- iii. $A-B-D$ forms an immorality, so conditioning on B makes A and D dependent on each other
- iv. Paths $C-D-B-E$ and $C-A-B-E$ are blocked by B

Thus ii and iv are correct

(b) Find $\Pr(A = 0, B = 1, C = 0, D = 1)$

..1

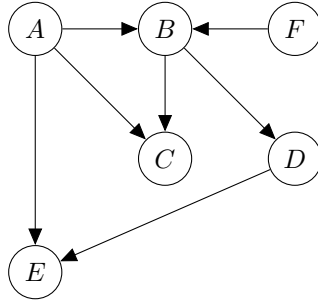
$\Pr(A, B, C, D) = \Pr(A) \Pr(B \mid A, D) \Pr(C \mid A) \Pr(D \mid C)$. On solving, the obtained answer is 0.096.

(c) Find $\Pr(E = 1 \mid B = 0, C = 1, D = 1)$

..1

$\Pr(E \mid B, C, D) = \Pr(E \mid B)$ as $E \perp\!\!\!\perp C, D \mid B$. From the given data $\Pr(E = 1 \mid B = 0) = 0.3$.

2. Given a BN:-



(a) Which of the following CIs is/are correct?

- i. $B \perp\!\!\!\perp E \mid D$
- ii. $A \perp\!\!\!\perp F \mid B$
- iii. $A \perp\!\!\!\perp D \mid B, C$
- iv. $A \perp\!\!\!\perp D \mid B, E$

..2 (multicorrect, -0.5 for each wrong answer)

- i. The path $E-A-B$ is not blocked
- ii. $A-B-F$ forms an immorality, so conditioning on B makes A and F dependent on each other
- iii. The path $A-B-D$ is blocked by B
- iv. $A-E-D$ forms an immorality, so conditioning on E makes A and D dependent on each other

Thus iii is correct

(b) Given the nodes B, E , how many nodes from the set $\{C, D, F\}$ are independent of the node A ? ..1

The answer is 0. $A-B-F$ and $A-E-D$ are immoralities conditioning on B, E makes A dependent on F and D . A is clearly dependent on C as they are connected by an edge.

(c) Which of these are local CIs?

- i. $A \perp\!\!\!\perp D$
- ii. $F \perp\!\!\!\perp A \mid C$
- iii. $C \perp\!\!\!\perp F \mid A, B$
- iv. $C \perp\!\!\!\perp E \mid B, D$

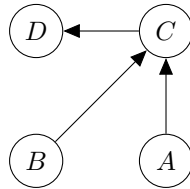
..2 (multicorrect, -0.5 for each wrong answer)

- i. $A \not\perp\!\!\!\perp D$ as there is a path $A-B-D$
- ii. $A-B-F$ forms an immorality, and $C \in \text{Desc}(B)$ so A and F are dependent on each other
- iii. $\text{Pa}(C) = \{A, B\}$ and $F \in \text{ND}(C)$
- iv. The path $C-A-E$ is not blocked

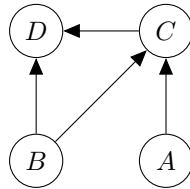
Thus iii is correct

3. Construct the BN from the following CIs with the order of the nodes being A, B, C, D using the subset construction method.

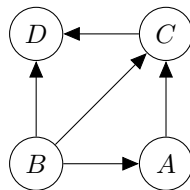
- $A \perp\!\!\!\perp B$
- $D \perp\!\!\!\perp A \mid C$



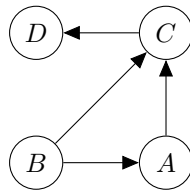
(a)



(b)



(c)



(d)

..1

In every step of the algorithm, $x_i \perp\!\!\!\perp Q_i - S \mid S$ according to the slides are-

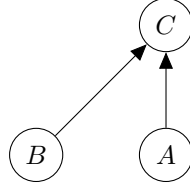
Step 1: $A \perp\!\!\!\perp \phi \mid \phi$



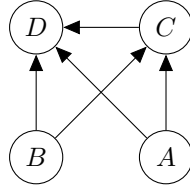
Step 2: $B \perp\!\!\!\perp A \mid \phi$



Step 3: $C \perp\!\!\!\perp \phi \mid A, B$



Step 4: $D \perp\!\!\!\perp \phi \mid A, B, C$



So, none of the options are correct.

4. Which of the following statements are correct?

- (a) Local CIs are a subset of Global CIs.
- (b) Local CIs can be derived through d-separation.
- (c) If $A \perp\!\!\!\perp B \mid P$, then $A \perp\!\!\!\perp B \mid Q$ if $P \subseteq Q$, where A, B, P, Q are sets of nodes.
- (d) If two variables A and B are conditionally independent given C in a BN, then they are d-separated by C in the same BN.

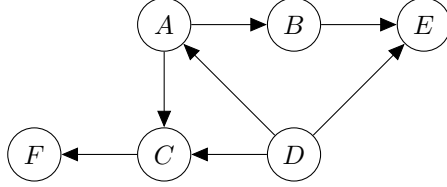
..2 (multicorrect, -0.5 for each wrong answer)

- (a) According to Theorem 3.4 in Koller-Freidman, it can be said that if X and Y are independent given Z in some distribution P that factorizes over G , then X and Y are d-separated given Z in G . So, Local CIs $\subseteq$ Global CIs.
- (b) Same as (a)
- (c) A counterexample can be shown. Let us consider a node $X, X \notin P, X \in Q$. Suppose $A-X-B$ forms an immorality, so $A \not\perp\!\!\!\perp B \mid Q$.
- (d) According to Theorem 3.3 in Koller-Freidman, $\mathcal{I}(G) \subseteq \mathcal{I}(P)$, and by definition 3.8, if $(X \perp\!\!\!\perp Y \mid Z) \in \mathcal{I}(P)$, then $\text{dsep}_G(X; Y \mid Z)$. So every independence in $\mathcal{I}(G)$ are d-separations.

Thus (a), (b) and (d) are correct.

5. Given each node can take values from the set $\{0, 1, 2\}$. Find the number of free parameters for:-

(Free Parameters Example: Let X be a discrete random variable with probability distribution $\Pr(X)$, where X can take values in the set $\{0, 1, 2\}$. Although there are three possible outcomes, only **two free parameters** are required to fully specify $\Pr(X)$, because $\Pr(X = 2) = 1 - \Pr(X = 0) - \Pr(X = 1)$)



- (a) The joint distribution A,B,C,D,E,F without the BN ..1

$\Pr(A, B, C, D, E, F) = \Pr(A) \Pr(B | A) \Pr(C | A, B) \Pr(D | A, B, C) \Pr(E | A, B, C, D) \Pr(F | A, B, C, D, E)$ (according to chain rule of probability)

For a term $\Pr(X | Y_1, Y_2 \dots Y_n)$ the number of free parameters are $2 \cdot 3^n$.

This is because $\Pr(X = 0 | Y_1, Y_2, \dots, Y_n)$ and $\Pr(X = 1 | Y_1, Y_2, \dots, Y_n)$ can each be specified by 3^n parameters, which together determine $\Pr(X = 2 | Y_1, Y_2, \dots, Y_n)$, making its 3^n parameters dependent.

So the total number of free parameters are $2 \cdot 3^0 + 2 \cdot 3^0 + \dots + 2 \cdot 3^5 = 728$

- (b) The joint distribution A,B,C,D,E,F with the BN ..1

$\Pr(A, B, C, D, E, F) = \Pr(A | D) \Pr(B | A) \Pr(C | A, D) \Pr(D) \Pr(E | B, D) \Pr(F | C)$

Thus, the total number of free parameters are $2 \cdot 3^1 + 2 \cdot 3^1 + 2 \cdot 3^2 + 2 \cdot 3^0 + 2 \cdot 3^2 + 2 \cdot 3^1 = 56$

Total: 14
